

Contact Sub-Riemannian Hybrid System (CSRHS)

Specification v0.2

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Abstract

This specification defines the Contact Sub-Riemannian Hybrid System (CSRHS), a mathematical structure that reconciles continuous Riemannian dynamics with discrete tick-based governance. Version 0.2 incorporates sixteen architectural improvements (C1–C8, D1–D8) discovered during prototype implementation in the governed geodesic controller and MaL-Cog compiler. Key additions include: Jacobi field monitoring for conjugate-point safety (C1), symplectic cost accumulation on the cotangent bundle (C2), sheaf-theoretic consistency verification across chart atlases (C3), Merkle DAG witness graphs with branching and pruning (C4), Fisher-Rao natural gradient projection (C5), genus-aware topological trajectory analysis (C6), exponential map integration replacing Euler steps (D1), holonomy-based damping via Wilson loop tracking (D2), Ollivier-Ricci/Wasserstein curvature replacing angle defect (D3), CG7 nine-axiom behavioral gating (D4), geodesic-aware horizontal flow with parallel transport (D5), forward-mode automatic differentiation replacing finite differences (D6), LRU geometry caching (D7), and tiered early-exit verification (D8). The liquid time constant is extended to a six-factor modulation function. All v0.1 structures, theorems, and interfaces are retained as backward-compatible special cases.

1. Motivation

1.1 The Tension (unchanged from v0.1)

Two paradigms exist for AI system dynamics: discrete ticks (deterministic replay, archive sufficiency, clear governance boundaries; but fixed granularity) and continuous ODEs (adaptive compute, geometric expressivity; but solver nondeterminism and unclear verification points). CSRHS reconciles these via a liquid time constant derived from manifold geometry that induces discrete tick boundaries while preserving the expressivity of continuous horizontal flows.

1.2 Core Insight (unchanged from v0.1)

(1) Continuous dynamics live in the horizontal distribution of a sub-Riemannian manifold. (2) Discrete governance occurs at tick boundaries induced by a liquid time constant. (3) Witness-based verification certifies reachability without computing continuous infima.

1.3 Implementation-Discovered Gaps [NEW]

Prototype implementation of v0.1 in the governed geodesic controller (Rust, 2200+ lines) and MaL-Cog compiler (1200+ lines) revealed sixteen architectural gaps. These fall into two categories:

Structural gaps (C1–C8) — the v0.1 specification was correct but underspecified. The implementation required concrete decisions about: geodesic deviation monitoring (C1), phase-space cost accounting (C2), multi-chart consistency (C3), non-linear witness structure (C4), information-geometric projection (C5), manifold topology awareness (C6), second-order geometry in the time constant (C7), and parallel branch exploration (C8).

Computational gaps (D1–D8) — the v0.1 specification assumed capabilities (exponential map, intrinsic curvature, exact gradients) without specifying how they are computed. The implementation required concrete algorithms for: manifold-respecting state advancement (D1), holonomy measurement (D2), optimal-transport curvature (D3), behavioral prediction (D4), connection-aware flow (D5), automatic differentiation (D6), result caching (D7), and verification tiering (D8).

2. Mathematical Framework

2.1 The CSRHS Tuple (extended)

Definition 2.1 (Contact Sub-Riemannian Hybrid System, v0.2):

A CSRHS is a tuple:

$$H = (M, g, D, \tau, \Pi, G, \Gamma, \text{Exp}, \nabla, A, P)$$

where the first seven components are as in v0.1, and the new components are:

Component	Type	Description	Source
Exp	Map	Exponential map $\text{Exp}: TM \rightarrow M$	D1
∇	Connection	Levi-Civita or Fisher-Rao connection	D2, D5
A	Atlas	Chart atlas $\{(U_a, \phi_a)\}$ covering M	C3
P	Predictor	CG7 behavioral gate $P: M \rightarrow \{\text{Admit}, \text{Reject}\}$	D4

The v0.1 components (M, g, D, τ , Π , G, Γ) are unchanged. Backward compatibility: setting Exp = additive Euler step, ∇ = flat connection, A = single global chart, and P = trivially admitting recovers the v0.1 tuple exactly.

2.2 The Horizontal Distribution (unchanged)

Following SR_Genealogy v1.2, the horizontal distribution at x in M is $D_x = \text{Mask}(x) \cap L(x)$, with the row-conservation constraint: for all rows r, the sum of $\Delta[r, c] = 0$ over columns c in $\text{Mask}(x, r)$. This is unchanged from v0.1.

2.3 The Carnot-Caratheodory Metric (unchanged)

$d_{CC}(x, y) = \inf$ of the integral of $|\dot{\gamma}(t)|_g dt$ over horizontal paths γ from x to y. We do NOT compute d_{CC} directly; we verify bounded witnesses. Unchanged from v0.1.

2.4 The Liquid Time Constant (extended) [CHANGED]

v0.2: Six-factor modulation replacing three-factor

Definition 2.3' (Liquid Time Constant, v0.2):

$$\tau(x, I) = \tau_{\text{base}} * \Phi_v1(\kappa, \nabla V, \sigma) * J(x) * W(x) * K(x) * H(x)$$

where Φ_v1 is the v0.1 modulation function and the four new factors are:

J(x) = Jacobi damping factor (C1). Equals 1.0 when the Jacobi field is healthy; approaches 0 near conjugate points where the exponential map degenerates.

W(x) = Holonomy damping factor (D2). Equals $\exp(-k * \text{severity}^2)$ where $\text{severity} = \|\text{Wilson loop} - I\|_F$ over a sliding window of recent ticks. Damps τ when parallel transport around the

trajectory fails to return to identity.

K(x) = Curvature gradient damping (C7). Equals $1/(1 + \delta * |\nabla \kappa|)$. Approaching a high-curvature region demands caution even when current curvature is moderate.

H(x) = Hessian spread damping (C7). Equals $1/(1 + \eta * \text{spread}(\text{Hess } V))$. Near saddle points of the value function, the eigenvalue spread of the Hessian is large, signaling directional instability.

The clamping constraint $\tau_{\min} \leq \tau \leq \tau_{\max}$ is preserved. With $J = W = K = H = 1$ (flat manifold, no holonomy, constant curvature, no saddle), this reduces exactly to the v0.1 formula.

2.5 State Advancement via Exponential Map [NEW]

v0.2 addition (D1): replaces $x + \Delta$ Euler step

Definition 2.5 (Exponential Map Advancement):

State advancement from x_i by step Δ_{i+1} uses:

$$x_{\{i+1\}} = \text{Exp}_x(\alpha * \Delta)$$

Three strategies are specified:

Retraction: $x + \alpha * \Delta$, then project to manifold. $O(n^2)$. For stochastic matrices: clamp negatives to zero, renormalize rows.

Lie exponential: $x * \exp(\alpha * \Delta)$ via matrix exponential. $O(n^3)$. Exact on $GL(n)$; preserves stochasticity when Δ has zero row sums.

Taylor approximation: truncated power series with controllable accuracy. $O(N * n^3)$ for N terms.

The Lie exponential is the correct geodesic step for the stochastic matrix manifold. The key property: when Δ is horizontal (zero row sums, per Section 2.2), $\exp(\Delta)$ is itself a stochastic matrix, so $x * \exp(\Delta)$ remains on the manifold without projection. This eliminates the v0.1 problem where $x + \Delta$ could produce negative entries or rows not summing to 1.

Validity condition V2 is updated: V2' requires $x_i = \text{Exp}_{x_{\{i-1\}}}(\Delta_{i1})$ rather than $x_i = x_{i-1} + \Delta_{i1}$. Setting $\text{Exp} = \text{additive}$ recovers V2.

2.6 Curvature Computation (replaced) [CHANGED]

v0.2: Ollivier-Ricci/Wasserstein replaces angle defect (D3)

v0.1 Section 5.1 specified scalar curvature via angle defect: $\kappa(x) = \text{average of } (1 - \cos(\theta_{xy}))$ over neighbors. This measures ambient Euclidean curvature, not the intrinsic curvature of Theory Space as a statistical manifold.

Definition 2.6 (Ollivier-Ricci Curvature):

$$\kappa(x, y) = 1 - w_1(\mu_x, \mu_y) / d(x, y)$$

where W_1 is the 1-Wasserstein distance between random walk distributions μ_x and μ_y . For a stochastic matrix A , $\mu_x = A[x, :]$ (the x -th row). W_1 is computed via Sinkhorn-regularized optimal transport with entropic regularization $\epsilon = 0.1$ and 50 iterations.

Interpretation: Positive curvature means random walks starting at nearby points cluster (the manifold is "convergent"). Negative curvature means they diverge (the manifold is "expansive"). This captures the graph-theoretic structure of Theory Space rather than the ambient geometry.

The angle-defect approximation is retained as a fallback for computational environments where Sinkhorn is too expensive. The K-Obligations (Section 7.3) apply to whichever curvature is configured.

3. Discrete Witness Structure (extended)

3.1 Extended Tick Record [CHANGED]

v0.2: tick record extended with phase, Jacobi, and chart data

Definition 3.1' (CSRHS Tick Record, v0.2):

Each tick record contains the v0.1 fields (x_i , Δ_i , τ_i , κ_i , input_hash , tick_index , cost_accumulated) plus:

jacobi_deviation: Q32.32 — $\|J(t)\|$ at this tick (C1)

jacobi_status: {Safe, Contracting, ConjugateImminent} (C1)

momentum: Q32.32 vector — phase-space momentum p (C2)

energy: Q32.32 — Hamiltonian $H(q,p)$ (C2)

chart_id: u32 — which chart this tick was computed in (C3)

cg7_verdict: {Admit, Reject, Escalate} (D4)

3.2 DAG-Structured Witness Graph [NEW]

v0.2 addition (C4): replaces linear tick sequence

The v0.1 witness stores ticks as a linear sequence $\text{Vec}\langle\text{TickRecord}\rangle$. This prevents speculative execution (trying multiple flow directions in parallel) and makes rollback expensive (discard entire suffix).

Definition 3.2' (Tick DAG):

A Tick DAG is a directed acyclic graph where each node contains a tick record and Merkle hash. A node's hash = $\text{SHA-256}(\text{parent_hashes} \parallel \text{state_hash})$. Nodes can have multiple parents (merging) or multiple children (branching). Active tips are tracked for the executor.

Branch status: Active (live exploration), Pruned (CG7-rejected or topologically trapped, kept for audit), Merged (converged with another branch).

The $\text{linearize}(\text{tip_id})$ function extracts a linear path from genesis to any tip for v0.1 compatibility. The v0.1 witness structure is the special case of a DAG with no branching.

3.3 Extended Witness Validity [CHANGED]

v0.2 extends the validity conditions V1–V7 with V8–V12:

Cond.	Name	Description	Source
V1	Horizontal	Δ_i in $D_{\{x_{i-1}\}}$ [unchanged]	v0.1
V2'	Path consistency	$x_i = \text{Exp}_{\{x_{i-1}\}}(\Delta_i)$ [D1: uses exp map]	D1
V3	Tick bounded	$n \leq \text{budget}$ [unchanged]	v0.1
V4	Liquid cost	$\sum \ \Delta_i\ /\tau_i \leq \text{budget}$ [unchanged]	v0.1
V5	Governance	W_G, W_S, W_F valid at x_n [unchanged]	v0.1

Cond.	Name	Description	Source
V6	tau consistency	$\tau_i = \tau_v3(x_{i-1}, l_{i-1})$ [extended formula]	C7,D2
V7	kappa consistency	$\kappa_i = \kappa_W(x_{i-1})$ [Wasserstein kappa]	D3
V8	Sheaf consistency	Horizontal in ALL covering charts [NEW]	C3
V9	Jacobi safety	No tick at conjugate point [NEW]	C1
V10	Energy conservation	Hamiltonian drift $\leq 2 \times$ dissipation [NEW]	C2
V11	Topology	Pathological loops $\leq$ genus threshold [NEW]	C6
V12	DAG integrity	Merkle hashes verify, all reachable from genesis [NEW]	C4

3.4 Archive Sufficiency (extended)

Theorem 3.1 (Archive Sufficiency) extends to v0.2: all new fields (Jacobi deviation, momentum, energy, chart_id, CG7 verdict) are deterministic pure functions of the trajectory state and are therefore recomputable from the archived Q32.32 data. The DAG structure includes Merkle hashes that are self-verifying. No additional oracle attestations are required.

4. Hybrid Dynamics (extended)

4.1 Continuous Flow (extended) [CHANGED]

v0.2: geodesic flow replaces Euclidean flow (D5)

Between tick boundaries, the system evolves via geodesic flow on the manifold rather than Euclidean translation:

$$dx/dt = f(x, I) \text{ in } D_x$$

subject to:

- $f(x, I)$ in D_x (horizontal constraint) [unchanged]
- Integral of $ds/\tau < 1$ (within single tick) [unchanged]
- f projected via Fisher-Rao natural gradient when on statistical manifold [NEW, C5]
- State advanced via $\text{Exp}_x(\alpha * f \, dt)$ not $x + f \, dt$ [NEW, D1]
- f parallel-transported along the geodesic segment [NEW, D5]

4.2 Discrete Jump (extended) [CHANGED]

At tick boundaries, the system now performs:

```
Jump_v2(x, t) = {  
  cg7_gate: assert P(x) = Admit      [NEW, D4]  
  commit: worldline.append(x, t)      [unchanged]  
  verify: assert BoxG and BoxS and BoxF [unchanged]  
  sheaf: assert consistent across atlas [NEW, C3]  
  topology: check pathological loops   [NEW, C6]  
  advance: tick := tick + 1            [unchanged]  
  homotopy: if drift > delta then trigger [unchanged]  
}
```

The CG7 behavioral gate (D4) runs BEFORE commit. A state that passes modality checks but fails CG7 is behaviorally unsafe; the branch is pruned in the DAG rather than aborting the entire execution.

4.3 Tick Guard Conditions (extended) [CHANGED]

v0.2: symplectic cost replaces displacement cost (C2)

v0.1: a tick fires when accumulated cost = $\| \Delta \| / \tau$ reaches 1.

v0.2: when symplectic mode is enabled (C2), the cost includes kinetic energy:

$$\text{cost_v2} = \| \Delta \| / \tau + (1/2) \| p \|^2 / \tau$$

where p is the momentum covector from the symplectic accumulator. This makes kinetic energy visible in the tick guard: a trajectory that decelerates sharply wastes energy, which now triggers an earlier tick (and hence earlier governance verification).

4.4 Speculative Execution [NEW]

v0.2 addition (C8): parallel branch exploration

At each tick, the executor may propose MULTIPLE flow directions. Each direction produces a tentative tick node in the DAG (C4). After CG7 gating (D4) and sheaf verification (C3), the best branch is kept and the rest are pruned. This is controlled by the `speculative_branches` configuration parameter (default: 3).

5. Curvature and Gradient Computation (rewritten)

5.1 Wasserstein Curvature [CHANGED]

See Section 2.6 for the Ollivier-Ricci curvature definition (D3). The Sinkhorn algorithm computes an entropic-regularized optimal transport plan in $O(n^2 * \text{iterations})$, replacing the v0.1 angle defect approximation.

5.2 Value Gradient via Automatic Differentiation [CHANGED]

v0.2: AD replaces finite differences (D6)

v0.1 computed gradients via central finite differences: $\partial f / \partial x_i$ approximately equals $(f(x + \epsilon e_i) - f(x - \epsilon e_i)) / 2\epsilon$. This requires $2 * \text{dim}$ function evaluations per gradient and is sensitive to the choice of epsilon (catastrophic cancellation in Q32.32 at small epsilon, truncation error at large epsilon).

v0.2 specifies forward-mode automatic differentiation using dual numbers. A dual number $x + x' * \epsilon$ carries both value and derivative; arithmetic operations propagate derivatives exactly. For Q32.32, dual numbers maintain the same precision as the primal computation. The AD interface replaces `compute_value_gradient`, `compute_curvature_gradient`, and `compute_hessian_spread`. Finite differences are retained as a fallback.

5.3 Jacobi Field Evolution [NEW]

v0.2 addition (C1): conjugate point detection

Definition 5.3 (Jacobi Field Monitor):

The Jacobi equation $\nabla^2 J / dt^2 + R(J, \gamma') \gamma' = 0$ is integrated via Stormer-Verlet leapfrog alongside the tick sequence. The Jacobi field $J(t)$ measures geodesic deviation: when $\|J\|$ collapses toward zero, nearby geodesics are focusing toward a conjugate point.

Status classification:

- **Safe:** $\|J\| > \text{threshold}$
- **Contracting:** $\|J\| > \text{threshold}$ but $d\|J\|/dt < 0$
- **ConjugateImminent:** $\|J\| < \text{threshold}$

At ConjugateImminent, the executor escapes via perturbation orthogonal to the current velocity (Gram-Schmidt against the Jacobi field direction). Validity condition V9 ensures no tick is committed at a conjugate point.

5.4 Holonomy Measurement [NEW]

v0.2 addition (D2): Wilson loop severity

A Wilson loop tracker maintains a sliding window of recent ticks (default: 16). The holonomy W is the parallel transport around the window: $W = \text{product of } T(x_i, x_{i+1})$ for consecutive ticks, where the transport matrix uses information-geometric scaling $T_{ij} = \sqrt{(\text{to}_{ij} / \text{from}_{ij})}$. Severity = $\|W - I\|_F$. When severity exceeds a threshold (default: 0.12), the holonomy damping factor $W(x)$ in the tau formula activates.

5.5 Geometry Caching [NEW]

v0.2 addition (D7): LRU cache for pure computations

Geometry computations (curvature, gradient, stationary distribution, Hessian spread, curvature gradient) are pure functions of the point. An LRU cache (default: 256 entries) keyed on point hash eliminates redundant recomputation across tau calculation, tick record construction, and verification. Cache consistency is guaranteed by purity: the invariant `cache.get(x) == compute(x)` holds by construction.

6. Behavioral Gate and Topology [NEW]

6.1 CG7 Nine-Axiom Behavioral Gate

v0.2 addition (D4): behavioral safety before tick commit

The modality checks (BoxG, BoxS, BoxF) verify STRUCTURAL validity. The CG7 predictor verifies BEHAVIORAL safety: whether the proposed state is within the admissible region of the behavioral prediction model.

Axiom	Name	Check	Complexity
1	Lipschitz	$ \Delta _{\infty} \leq \text{bound}$	$O(n)$
2	Drift	Weighted L2 from baseline $\leq$ limit	$O(n)$
3	Spectral	$\rho(A) < 1 - \text{margin}$	$O(n^2)$
4	Row conservation	All rows sum to 1	$O(n^2)$
5	Positivity	All entries ≥ 0	$O(n^2)$
6	Connectivity	Transition graph connected	$O(n+m)$
7	Sign structure	Domain-specific sign constraints	$O(n^2)$
8	Identity exclusion	Distance from identity $>$ radius	$O(n^2)$
9	Sentinel invariance	Probes at test points stable	$O(S *n)$

CG7 verdict: Admit (proceed), Reject (prune branch in DAG), Escalate (log warning, proceed with caution). Rejection prunes the DAG branch rather than aborting the entire execution, enabling the speculative execution pattern (Section 4.4).

6.2 Genus-Aware Topological Analysis

v0.2 addition (C6): structural vs pathological loop classification

The tick topology analyzer computes approximate persistent homology on the trajectory point cloud. Detected H_1 features (loops) are classified as:

Class	Criterion	Action
Structural	Count $\leq \beta_1 = 2g$ AND matches handle circumference	Continue (expected from manifold genus)
Pathological	Count $> \beta_1$ OR excessive winding OR micro-oscillation	BreakLoop or abort
Ambiguous	Within budget but no circumference match	Continue with monitoring

The abort threshold is $\text{pathological_loops} > \max(1, \text{ceil}(\beta_1/2))$. For genus 0 (sphere): threshold = 1. For genus 3 (Klein's quartic): threshold = 3. For genus 14 (Hurwitz triplet): threshold = 14. The v0.1 system implicitly assumed genus 0 (any loop is pathological).

The ManifoldTopology descriptor can be constructed from: explicit genus, simplicial data ($V - E + F = \chi = 2 - 2g$), or Gauss-Bonnet integral ($\int K \, dA = 2\pi\chi$). For genus ≥ 6 , sheaf checking (C3) is recommended as mandatory, based on polyhedral embedding research showing that embeddability is not guaranteed.

6.3 Sheaf-Theoretic Consistency

v0.2 addition (C3): multi-chart verification

v0.1 checks horizontal membership at a single chart. When the manifold requires multiple charts (atlas A), a step Delta that is horizontal in chart alpha may NOT be horizontal in chart beta. This is "governance arbitrage" across the atlas.

v0.2 evaluates horizontal membership in ALL covering charts and enforces the gluing condition: all charts must agree. Verdicts: Consistent (all agree), Inconsistent (chart disagreement — surfaces as V8 failure), SingleChart (only one chart covers the point).

7. Obligation Test Suite (extended)

v0.1 Sections 7.1–7.4 are retained unchanged. v0.2 adds Sections 7.5–7.8.

7.1–7.4 (unchanged from v0.1)

H-Obligations (H1–H4), T-Obligations (T1–T4), K-Obligations (K1–K3), and G/S/F-Obligations (G1–G4, S1–S4, F1–F5) are unchanged.

7.5 Jacobi Safety Obligations (J-Obligations) [NEW]

J1: For all ticks i , $\text{jacobi_status} \neq \text{ConjugateImminent}$.
J2: Jacobi deviation is monotonically tracked (norm_history maintained).
J3: Escape perturbation applied only when ConjugateImminent detected.

7.6 Phase Space Obligations (P-Obligations) [NEW]

P1: Momentum norm bounded: $\|p\| \leq \text{max_momentum}$.
P2: Energy drift bounded: $|H_k - H_0| \leq 2 * n * \text{friction}$.
P3: Symplectic form preservation: $|\omega(\text{dp}, \text{dq}) - \omega_0| < \epsilon$.

7.7 Sheaf Obligations (Sh-Obligations) [NEW]

Sh1: For all ticks, horizontal check consistent across covering charts.
Sh2: No chart disagreement on any tick.
Sh3: For genus ≥ 6 manifolds, sheaf checking is mandatory (not optional).

7.8 Behavioral Obligations (B-Obligations) [NEW]

B1: CG7 verdict = Admit for every committed tick (when predictor enabled).
B2: DAG Merkle hashes verify for all nodes.
B3: All DAG nodes reachable from genesis.
B4: Pruned branches retain audit trail.

7.9 Topological Obligations (Top-Obligations) [NEW]

Top1: Pathological loops $\leq \max(1, \text{ceil}(\beta_1/2))$.
Top2: Structural loops correctly classified against handle circumferences.
Top3: Genus estimate consistent with simplicial or Gauss-Bonnet data.

8. Canonical Encoding (extended)

8.1 CSRHS Tick Record v0.2

```
{ "schema": "CSRHS_TICK_v0.2",  
  "x": "<Q32.32 vector>",  
  "delta": "<Q32.32 vector>",  
  "tau": "<Q32.32>",  
  "kappa": "<Q32.32>",  
  "input_hash": "<SHA-256>",  
  "tick_index": "<u64>",  
  "cost_accumulated": "<Q32.32>",  
  "jacobi_deviation": "<Q32.32>",  
  "jacobi_status": "<Safe|Contracting|ConjugateImminent>",  
  "momentum": "<Q32.32 vector>",  
  "energy": "<Q32.32>",  
  "chart_id": "<u32>",  
  "cg7_verdict": "<Admit|Reject|Escalate>" }
```

8.2 CSRHS Witness Bundle v0.2

```
{ "schema": "CSRHS_WITNESS_v0.2",  
  "x0": "<starting point>",  
  "xn": "<ending point>",  
  "ticks": [ "<CSRHS_TICK_v0.2>", ... ],  
  "tick_dag": "<optional DAG structure>",  
  "budget": "<u64>",  
  "total_cost": "<Q32.32>",  
  "modality_witnesses": { "G": "...", "S": "...", "F": "..."},  
  "content_hash": "<SHA-256>",  
  "topology_diagnosis": "<optional>",  
  "energy_drift": "<Q32.32>",  
  "dag_tip": "<optional u64>" }
```

8.3 Hash Computation (unchanged)

content_hash = SHA-256(JCS_canonicalize(witness_without_hash)). Per RFC 8785 for JSON Canonicalization Scheme. For DAG nodes, merkle_hash = SHA-256(parent_hashes || state_hash).

9. Tiered Verification [NEW]

v0.2 addition (D8): early-exit verification tiers

v0.1 runs all checks sequentially. v0.2 introduces tiered verification with four levels, each short-circuiting on failure:

Tier	Complexity	Checks	Use Case
0: Fast	$O(1)$	Content hash, budget, delta magnitude	Preview / fast reject
1: Behavioral	$O(n)$	CG7 gate, Jacobi safety, energy drift	Behavioral pre-screen
2: Structural	$O(n \cdot \text{dim})$	Horizontal, path, liquid time, modalities	Standard verification
3: Full	$O(n^2)$	Sheaf, topology, geometry, DAG integrity	Production commit

Callers can request a specific tier. Speculative branches use Tier 1 (fast behavioral rejection) during exploration, then Tier 3 for the selected branch before promotion.

10. Theorems and Proofs (extended)

10.1 Soundness Theorem (extended)

Theorem 10.1' (CSRHS v0.2 Soundness):

If witness W satisfies V1–V12, then there exists a horizontal path $\gamma: [0, T] \rightarrow M$ with: (1) $\gamma(0) = x_0$, $\gamma(T) = x_n$; (2) $\gamma'(t)$ in $D_{\gamma(t)}$ for all t ; (3) integral of $ds/\tau \leq$ budget; (4) γ is governance-admissible; (5) γ stays on the manifold (no off-manifold excursions); (6) γ avoids conjugate points; (7) γ is behaviorally admissible (CG7).

Proof sketch: Conditions (1)–(4) follow from v0.1 proof with Exp replacing additive step (the exponential map preserves horizontality for horizontal initial conditions). (5) follows from V2' and the properties of the Lie exponential on the stochastic matrix manifold. (6) follows from V9 and the Jacobi field monitor. (7) follows from V8 and the CG7 predictor applied at every tick. QED

10.2 Completeness Theorem (unchanged)

Theorem 10.2 (CSRHS Completeness) is unchanged: any governance-admissible horizontal path with finite liquid cost can be discretized into a valid CSRHS witness. The v0.2 extended fields are computed from the path.

10.3 Archive Sufficiency (extended)

Theorem 10.3 (Tier=Full Eliminability) extends to v0.2: all new fields are deterministic pure functions of the trajectory, recomputable from Q32.32 archived data. The DAG's Merkle structure is self-verifying. CG7 verdicts are deterministic given the predictor configuration (which must also be archived).

10.4 Manifold Membership Theorem [NEW]

Theorem 10.4 (Manifold Membership):

If the exponential map strategy is LieExponential and all steps are horizontal (V1 satisfied), then x_i is a stochastic matrix for all i .

Proof: For a stochastic matrix A and horizontal vector Δ (zero row sums), $\exp(\Delta)$ has rows summing to 1 (because $\exp(M) * 1 = 1$ when $M * 1 = 0$, where 1 is the all-ones vector). Therefore $A * \exp(\Delta)$ is the product of two stochastic matrices, which is stochastic. QED

11. Safety Analysis (extended)

11.1 Extended Failure Modes

Failure Mode	Detection	Mitigation	Source
$\tau \rightarrow 0$	T4: $\tau \geq \tau_{\min}$	Fail-closed	v0.1
$\kappa \rightarrow \infty$	K2: κ bounded	Horizontal escape	v0.1
Budget exhaustion	V3, V4	Graceful degradation	v0.1
Conjugate point	V9: Jacobi monitor	Orthogonal escape (C1)	v0.2
Energy blowup	V10: Hamiltonian drift	Momentum clamp (C2)	v0.2
Chart disagreement	V8: Sheaf check	Witness rejection (C3)	v0.2
Topological trap	V11: Loop analysis	Perturbation/abort (C6)	v0.2
Behavioral drift	CG7 axioms 1-9	Branch pruning (D4)	v0.2
Off-manifold	V2': Exp map check	Re-projection (D1)	v0.2
Holonomy buildup	Wilson loop severity	τ damping (D2)	v0.2

11.2 Extended Attack Vectors

Attack	Defense	Source
Forge non-horizontal Delta	Mask/L verified + sheaf (C3)	v0.1 + v0.2
Manipulate tau	τ recomputed, 6-factor formula	v0.1 + v0.2
Exploit curvature	Wasserstein curvature (D3)	v0.2
Governance arbitrage across charts	Sheaf gluing condition (C3)	v0.2
Trigger false conjugate escape	Jacobi field is deterministic (C1)	v0.2
Hide behavioral drift	CG7 9-axiom gate (D4)	v0.2
Topological trap (orbit basin)	Genus-aware topology (C6)	v0.2
Cache poisoning	Cache keyed on hash; pure functions	v0.2

12. Parameter Selection (extended)

12.1 v0.1 Defaults (unchanged)

<code>tau_base = 1.0</code>	<code>alpha = 0.1</code>	<code>beta = 1.0</code>	<code>gamma = 0.05</code>
<code>tau_min = 0.01</code>	<code>tau_max = 10.0</code>	<code>kappa_max = 100.0</code>	<code>delta_kappa = 10.0</code>

12.2 v0.2 Additional Parameters [NEW]

Parameter	Default	Description
<code>jacobi_collapse</code>	0.01	Jacobi field collapse threshold (C1)
<code>symplectic_friction</code>	0.05	Dissipation rate (C2)
<code>max_momentum</code>	10.0	Momentum clamp (C2)
<code>fisher_rao_eps</code>	0.001	Fisher-Rao regularization (C5)
<code>topo_persistence</code>	0.5	Min loop persistence (C6)
<code>max_winding</code>	3	Winding number ceiling (C6)
<code>holonomy_threshold</code>	0.12	Wilson loop severity trigger (D2)
<code>holonomy_damping_k</code>	10.0	Gaussian damping steepness (D2)
<code>wilson_window</code>	16	Trajectory window for holonomy (D2)
<code>wasserstein_reg</code>	0.1	Sinkhorn regularization (D3)
<code>sinkhorn_iters</code>	50	Sinkhorn iterations (D3)
<code>cg7_lipschitz</code>	0.5	Axiom 1 bound (D4)
<code>cg7_drift_limit</code>	1.0	Axiom 2 limit (D4)
<code>cg7_spectral_margin</code>	0.05	Axiom 3 margin (D4)
<code>cg7_core_radius</code>	0.1	Axiom 8 radius (D4)
<code>exp_map_terms</code>	10	Taylor series terms (D1)
<code>cache_size</code>	256	LRU cache entries (D7)
<code>speculative_branches</code>	3	Parallel branches (C8)
<code>verify_tier</code>	Full	Default verification tier (D8)

13. Integration with Existing Architecture (extended)

13.1–13.3 (unchanged from v0.1)

Connections to SR_Genealogy v1.2, Timeful Spec, and lambda-MaL Type System are unchanged. Compatibility is maintained: any v0.1 witness is a valid v0.2 witness with constant tau, additive Exp, single chart, no CG7, and genus-0 topology.

13.4 Connection to Polyhedral Embedding Research [NEW]

The genus-aware topology module (C6) draws on Brehm-Kuhnel's work on polyhedral embeddings of higher-genus surfaces. Key integration points: embeddability boundary at genus 6 (sheaf checking mandatory), Klein's quartic (genus 3, 24 vertices) as a test fixture, symmetry reduction heuristic from Leonardo polyhedra, and the two-stage solver approach (broad search then refinement) mapping onto DAG branching.

13.5 Connection to Information Geometry [NEW]

The Fisher-Rao projector (C5), Wasserstein curvature (D3), and information-geometric parallel transport (D5) connect CSRHS to the Amari-Chentsov theory of statistical manifolds. The Fisher-Rao metric is the unique Riemannian metric invariant under sufficient statistics (Chentsov's theorem). The Ollivier-Ricci curvature extends this to discrete/graph settings via optimal transport.

14. Open Questions (updated)

v0.1 questions 1–5 are retained. v0.2 adds questions 6–12 based on implementation experience:

1. [v0.1] Optimal tau calibration: Can tau parameters be learned while maintaining safety?
2. [v0.1, PARTIALLY ADDRESSED] Higher-order curvature: Should tau depend on Riemann tensor beyond scalar curvature? **v0.2: Yes. C7 adds curvature gradient and Hessian spread. D2 adds integrated holonomy.**
3. [v0.1] Anisotropic tau: Should tau vary by direction in $T_x M$, not just by position?
4. [v0.1] Bracket-generating extensions: Can CSRHS support non-bracket-generating distributions via virtual ticks?
5. [v0.1] Quantum analogues: Is there a coherent extension to quantum state spaces?
6. [v0.2] Sinkhorn precision: What is the optimal entropic regularization for Q32.32 fixed-point Wasserstein computation?
7. [v0.2] CG7 axiom weighting: Should axioms have configurable severity (hard reject vs soft penalty)?
8. [v0.2] Genus estimation accuracy: How sensitive is the topology analyzer to errors in the genus estimate?
9. [v0.2] DAG pruning policy: What is the optimal balance between exploration breadth (branches) and commit latency?
10. [v0.2] AD in Q32.32: Can second-order (Hessian-vector products) AD be implemented efficiently in fixed-point arithmetic?
11. [v0.2] Holonomy vs topology: Are the Wilson loop tracker (D2) and the topology analyzer (C6) redundant, or do they capture genuinely different phenomena?
12. [v0.2] Sheaf relaxation: Should the sheaf checker allow approximate consistency (bounded disagreement) for large atlases?

15. Version History

Version	Date	Changes
0.1	2026-01-23	Initial specification
0.2	2026-02-07	C1-C8: Jacobi monitor, symplectic cost, sheaf verification, Merkle DAG, Fisher-Rao projection, genus- g

16. References

- [1] Devon. "SR_Genealogy_LambdaMaL v1.2." Production bundle, January 2026.
- [2] Devon. "Timeful Governed Self-Modification." Archive-sufficient report, January 2026.
- [3] Devon. "lambda-MaL Type System." Research paper, 2025.
- [4] Hasani, R. et al. "Liquid Time-constant Networks." AAAI 2021.
- [5] Montgomery, R. "A Tour of Subriemannian Geometries, Their Geodesics and Applications." AMS, 2002.
- [6] Devon. "MaL-Cortex Safety Architecture." Research paper, December 2025.
- [7] Devon. "Non-Forward-Closed Theory Spaces." Research report, 2025.
- [8] Devon. "governed_geodesic_controller_v2.rs." Rust prototype, February 2026. [NEW]
- [9] Devon. "genus_aware_topology.rs." Genus-calibrated topology module, February 2026. [NEW]
- [10] Devon. "csrhs_v03.malcog." MaL-Cog implementation v0.3, February 2026. [NEW]
- [11] Ollivier, Y. "Ricci curvature of Markov chains on metric spaces." J. Funct. Anal., 2009. [NEW]
- [12] Cuturi, M. "Sinkhorn Distances: Lightspeed Computation of Optimal Transport." NeurIPS, 2013. [NEW]
- [13] Amari, S. "Information Geometry and Its Applications." Springer, 2016. [NEW]
- [14] Brehm, U. and Kuhnel, W. "Polyhedral embeddings and immersions of triangulated manifolds." [NEW]
- [15] Devon. "Theory Spaces and Invariant Cores." Research paper, January 2026. [NEW]

Appendix B: Quick Reference Card (updated)

+-----+		
CSRHS v0.2 QUICK REFERENCE		
+-----+		
STRUCTURE: H = (M, g, D, tau, Pi, G, Gamma, Exp, nabla, A, P)		
LIQUID tau (v0.2):		
tau = tau_base * Phi * J * W * K * H		
Phi = 1/(1+a kappa +b nablaV +g*sigma) [v0.1]		
J = Jacobi damping [C1]		
W = Holonomy damping [D2]		
K = Curvature gradient damping [C7]		
H = Hessian spread damping [C7]		
TICK FIRES: when cost >= 1		
STEP: x_new = Exp_x(alpha * Delta) [D1]		
KAPPA: Ollivier-Ricci via Sinkhorn [D3]		
GATE: CG7 9-axiom behavioral [D4]		
VALID: V1-V12 (tiered D8)		
ARCHIVE: Full-tier eliminable, Q32.32		
+-----+		
DEFAULTS: tau_base=1, a=0.1, b=1.0, g=0.05		
tau_min=0.01, tau_max=10, kappa_max=100		
+ 20 v0.2 parameters (Section 12.2)		
+-----+		